Yash Diwan

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SUMMARY

Final-year B.Tech CSE student (8.69 CGPA) with hands-on experience building backend, full-stack, and Machine Learning applications across healthcare, real-time routing, and civic technology. Strong in Java, Data Structures and Algorithms, SQL, and core Computer Science fundamentals, with practical Python expertise in Data Science and Machine Learning. Skilled in Machine Learning, database systems, backend development, REST APIs, real-time systems, and AWS cloud foundations.

EDUCATION

VIT Bhopal University — Bhopal, India
BTech in CSE (CGPA: 8.69/10.00) — *2023 – 2027*

Shalom Hills International School — Gurgaon, India
CBSE Class XII (92.6%) | CBSE Class X (91.8%) — *2021 – 2023*

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL
CS Fundamentals: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks
Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, Exploratory Data Analysis (EDA), Machine Learning, Computer Vision, Generative AI
Backend & Web: Flask, FastAPI, REST APIs, WebSockets, Node.js
Databases: MySQL, MongoDB
Cloud & Tools: AWS, Git, GitHub, Render, Streamlit

PROJECTS

Crowd Manager: Autonomous Crowd Routing & Safety Control Room | github.com/yashdiwan12/Simhastha
Web & ML Engineering Project
Python, Scikit-learn, GoogleMaps API, Websockets, Fast API
- Developed a smart safety control room for Simhastha Kumbh Mela, the world's largest human gathering.
- Utilizing an A* routing engine that prevents stampedes with a low 10.85ms latency across a 5,000-node graph.
- Architected a prototype FastAPI backend broadcasting live updates every 2s.

HealBridge AI: Symptom-Based Disease Prediction System | github.com/yashdiwan12/Heal-BridgeAi
AI Engineering Project
Python, Flask, Scikit-learn, Pickle, HTML/CSS/JS
- Engineered a Flask-based web application utilizing machine learning pipelines for real-time disease diagnosis.
- Integrated ML pipelines to accurately predict diseases, achieving 97.62%+ accuracy on a dataset of 100+ symptoms.
- Reduced symptom entry time by 60% and improved UX by implementing a streamlined, categorized input system.

ImpACT: AI-Augmented Civic Action Platform | github.com/yashdiwan12/IMPACT
Data Science & ML Engineering Project
Python, Node.js, MongoDB, YOLO, React
- Developed a Pan-India web platform that automates civic complaint triage and municipal officer routing.
- Trained a custom YOLOv8 model over 100 epochs to detect civic hazards, achieving 0.89 precision on real-world citizen images.
- Built a Python/Flask ML microservice to serve real-time inferences, maintaining <450ms latency under 500 concurrent requests.

CERTIFICATIONS

- **IBM** – IBM Generative AI Engineering Certificate
- **ORACLE** – Oracle Database@AWS Certified Architect Professional Certificate
- **IBM** – Machine Learning with Python with Honors Certificate
- **AWS** - AWS Certified Solutions Architect – Associate Certificate

LEADERSHIP & EXTRACURRICULARS

Google Developer Groups VIT Bhopal | Core Team Member | Certificate — *Nov 2024 – Jun 2025*